

陈健怡

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About Me

我是一名新加坡南洋理工大学 (NTU) 的博士毕业年学生, 在洪伟德教授 (Prof. Ang Wei Tech) 指导下, 专注于辅助与康复设备中的物理人机交互仿真研究。我主要使用 MuJoCo 平台, 构建融合骨骼、肌肉骨骼与软体模型的数字人体, 并设计 Real2Sim2Real 框架, 实现个性化机器人控制策略。我的研究结合仿真驱动的机器学习, 旨在开发安全直观的交互策略, 推动智能物理交互系统的发展。目前, 我也在探索将基础模型与大语言模型 (Foundation & Large Language Models) 融入仿真中, 以提升人机交互的真实性、理解能力与泛化性。

Education

南洋理工大学, 新加坡 2021 年 9 月 – 2025 年 8 月(预计)
 博士 in 机械工程

- 导师: 洪伟德教授(Prof Ang Wei Tech)
- 拟定论文题目: 通过精确的人机闭环仿真, 加速辅助机器人的发展
- 研究方向: 基于物理的仿真、人机交互、机器学习、基础模型, 大语言模型

南洋理工大学, 新加坡 Aug 2017 – May 2021
 本科 in 机械工程 (机器人与机电一体化方向)

- 绩点: 4.84/5.00, 学年入选院长名单 (Dean's List) 2017/2018
- 奖项: 南洋奖学金 (Nanyang Scholarship)

Research Projects

Human-In-The-Loop Robotic Simulator Sept 2021 – present

A simulation framework for physical human-robot interaction (pHRI) investigation to personalise robotic systems.

- Developed a human-in-the-loop simulation framework for assistive robotics, featuring personalized digital twins with varying abilities and disabilities. The pipeline incorporates skeletal, musculoskeletal, and soft body models, using reinforcement learning to simulate diverse human capabilities and enable realistic, adaptive testing of robotic systems.
- Designed and validated a Real2Sim2Real framework to personalize robotic controllers based on individual user characteristics, enabling improved pHRI through simulation-informed adaptation and real-world testing.
- Investigating various human-robot interaction modalities in simulation, including soft body dynamics, mass-spring-damper models, and other methods to accurately represent physical interactions.
- Simulated a range of assistive robotic systems - including robot-assisted feeding arms, lower limb and upper limb exoskeletons, gait assistive robots, using MuJoCo.

Mobile Third Arm Robot 2020 – 2022

Vision-guided mobile third arm co-bot for patient transfers between bed and wheelchair.

- Developed a mobile third arm robot that assists caregivers for moderate assisted pivot transfer between the bed and wheelchair.
- Implemented a human-tracking computer vision algorithm to perform user following
- Developed human-robot interaction strategies for safe and intuitive operation of the mobile third arm robot between caregiver, patient and mobile third arm robot.

Design and Analysis of Underwater Robotic Systems 2018 – 2019

3D printed underwater robotic manipulator for object grasping.

- Designed and 3D printed a robotic manipulator for an autonomous underwater vehicle to grasp objects from the pool floor.
- Created a bio-inspired flexible hull structure that deforms with depth to enhance rigidity while reducing weight.

- Conducted finite element analysis to optimize both manipulator and hull designs for strength, hydrodynamic performance, and operational depth.

Work Experience

Project Officer, Rehabilitation Research Institute of Singapore – Singapore Aug 2021 – present

- Spearheaded the Human-In-The-Loop Robotic Simulator project, focusing on developing a platform that combines realistic human digital twins with assistive robots for human-in-the-loop simulations.
- Mentored seven undergraduate students on final year projects, guiding them through the research process and providing technical support.
- Assisted in preparing funding proposals to support current and future projects of the institute.

Engineering Intern, Oishii – New Jersey, USA June 2019 – June 2020

- Designed an indoor vertical farm structure accommodating 250,000 plants with a novel rack system that increases efficiency of construction and assembly process by 90%.
- Pioneered development of a USD 400,000 R&D facility consisting of ten independent growing environments tracking more than 80 plant parameters to determine optimal plant growth and yield.
- Led an engineering team of 12 to implement testing and troubleshooting schedule to all farm sub-systems as lead test engineer to bring the world's largest indoor strawberry vertical farm facility operational.
- Evaluated 32 engineering candidates for engineering roles, mentored 19 new engineering technicians to improve mechanical and electrical skillsets and align with company working norms.

Publications

ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework Oct 2025

Sherwin Stephen Chan[†], J Anne Yow[†], Yi Heng San, Vasanthamaran Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), [†] Denotes equal contribution

A Human-In-The-Loop Simulation Framework for Evaluating Control Strategies in Gait Assistive Robots May 2025

Yifan Wang[†], *Sherwin Stephen Chan*[†], Mingyuan Lei, Lek Syn Lim, Henry Johan, Bingran Zuo, Wei Tech Ang

IEEE International Conference on Robotics and Automation (ICRA), [†] Denotes equal contribution

Simulating Safe Bite Transfer in Robot-Assisted Feeding with a Soft Head and Articulated Jaw May 2025

Yi Heng San[†], Vasanthamaran Ravichandram[†], J Anne Yow, *Sherwin Stephen Chan*, Yifan Wang, Wei Tech Ang

IEEE International Conference on Rehabilitation Robotics (ICORR), [†] Denotes equal contribution

Personalised 3D Human Digital Twin with Soft-Body Feet for Walking Simulation Dec 2024

Kum Yew Loke, *Sherwin Stephen Chan*, Mingyuan Lei, Henry Johan, Bingran Zuo, Wei Tech Ang
International Conference on Social Robotics

Creation and evaluation of human models with varied walking ability from motion capture for assistive device development Sept 2023

Sherwin Stephen Chan, Mingyuan Lei, Henry Johan, Wei Tech Ang

IEEE International Conference on Rehabilitation Robotics (ICORR)

Investigation of Modeling Differences between OpenSim and Visual3D for Gait Analysis of Healthy Gait Aug 2023

Beth Eng Wan Xuan, *Sherwin Stephen Chan*, Henry Johan, Lek Syn Lim, Bingran Zuo, Wei Tech Ang
International Convention on Rehabilitation Engineering and Assistive Technology

Skills

Programming Languages: Python, C++, C

Robotics: MuJoCo, OpenSim, Machine Learning, Reinforcement Learning, IsaacLabs, ROS, SolidWorks, SolidEdge, STM32, ESP32, Linux

Languages: English, Chinese, Hokkien (Dialect), Tagalog